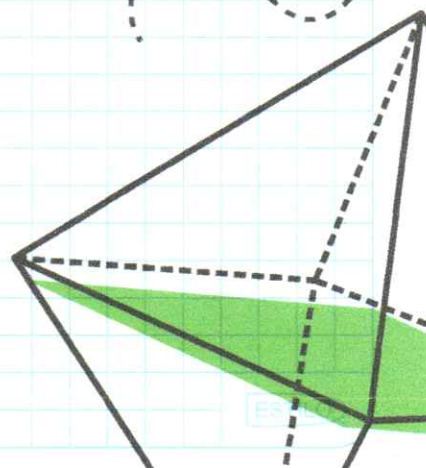
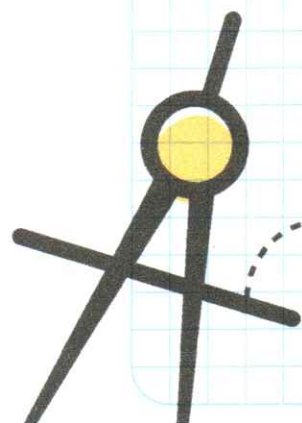
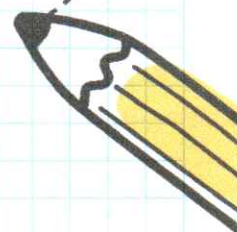
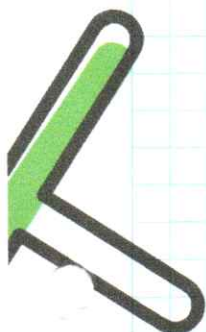
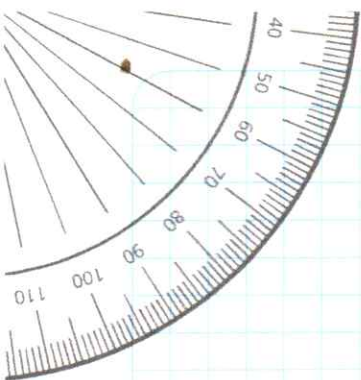


π

Pruebas

MATEMÁTICA III





Datos Informativos

Nombre del estudiante: Loor Marvin - 120741566-0

Docente: Mrs. Diego Tipán

Asignatura: Matemática III

Curso: Intercala

Tema: _____

Fecha: 08-09-26

3

$$1. f(x) = \sqrt[4]{x^5 + 5}$$

$$f(x) = \sqrt[4]{x^5} + \sqrt[4]{5}$$

$$f(x) = x^{5/4} + \frac{5^{1/4}}{4}$$

$$f'(x) = \frac{5}{4}x^{1/4} = \frac{1}{4}5^{-3/4}$$

$$f(x) = \sqrt[4]{\frac{5}{4}x} + \sqrt[4]{\frac{5}{4}}$$

$$f(x) = \sqrt[4]{\frac{5}{4}x} + \sqrt[4]{\frac{5}{4}}$$

$$f(x) = \sqrt[4]{\frac{5}{4}x} + \sqrt[4]{\frac{5}{4}}$$

$$f(x) = \sqrt[4]{\frac{5}{4}x} + \frac{5}{64}$$

15-sep-2026

$$2. f(x) = \frac{6x^4 + 7}{\sqrt{x^2 - 1}}$$

$$f(x) = \frac{uv^2 - uv^2}{v^2}$$

48
48
48
24
48

$$f(x) = \frac{(6x^4 + 7)(x - 1^{1/2})^2 - (x - 1^{1/2})(6x^4 + 7)^2}{(x - 1^{1/2})^2}$$

$$f(x) = \frac{(6x^4 + 7)(1 - \frac{3}{2}) - (x - 1^{1/2})(-24x^3)}{(x - 1^{1/2})^2}$$

$$f(x) = \frac{3 - 9}{2} = -3$$

$$f(x) = \frac{3 - 9}{2} = -3$$

$$f(x) = \frac{3 - 9}{2} = -3$$



Datos Informativos

Nombre del estudiante: Loay Tuzier Moira Aldana 12011966-0

Docente: Msc. Diego Tipán

Asignatura: Matemática III

Curso: Interciclo

Tema: Evaluación Unidad 1

Fecha: 17-09-2026

① $F(x) = \sin^2(5x+2)$. Explique y desarrolle el concepto de primitiva

$F'(x) = f(x)$. La antiderivada de $f(x)$ es igual a $F'(x)$, es decir, si integramos $f(x)$ nos dará $F(x)$, si derivamos $F(x)$ nos dará la antiderivada. Ejemplo:

$$f(x) = \sin^2(5x+2)$$

$$u = 5x+2 \\ du = 5dx$$

$$v = \sin u \\ dv = \cos u dx$$

$$= \int \sin^2(u) \frac{du}{5}$$

$$= \frac{1}{5} \cos u + C$$

$$= \int (\sin u)^2 \frac{du}{5}$$

$$F(x) = \frac{1}{5} \cos(5x+2) + C$$

$$= \int v^2 \frac{dv}{5}$$

$$F'(x) = f(x)$$

$$= \frac{1}{5} \int v^2 dv$$

② $\int \frac{3x^{1/2} + x^{3/2} + 3x^2}{\sqrt{x}} dx$. Resuelva la integral

$$= \int \frac{3x^{1/2}}{\sqrt{x}} dx + \int \frac{x^{3/2}}{\sqrt{x}} dx + \int \frac{3x^2}{\sqrt{x}} dx$$

$$= 3x + x\sqrt{x^2}$$

$$= 3 \int dx + \int x^2 dx + 3 \int x^{3/2} dx$$

$$= 3x + x \cdot x + \frac{6x^{5/2}}{5} + C$$

$$= 3 \int dx + \int x^2 dx + 3 \int x^{3/2} dx$$

$$= 3x + x^2 + \frac{6\sqrt{x^3}}{5} + C$$

$$= 3x + \sqrt{x^3} + \frac{3x^{3/2+1}}{\frac{3}{2}+1} + C$$

$$= 3x + x\sqrt{x^2} + \frac{3x^{5/2}}{5} + C$$

$$= 3x + x\sqrt{x^2} + \frac{6x^{5/2}}{5} + C$$

$$\textcircled{3} \int \sec x (\tan x + 1) dx$$

$$= \int \sec x (u) dx$$

$$= \int \sec x (u) du$$

$$= \int u du$$

$$= \frac{u^{1+1}}{1+1} + C$$

$$= \frac{u^2}{2} + C$$

$$= \frac{\sec^2 x (u)}{2} + C$$

$$u = \tan x + 1$$

$$du = \sec^2 x dx$$

$$v = \sec x (u)$$

$$dv = \sec x \tan x dx$$

$$= \frac{\sec^2 x (\tan x + 1)}{2} + C$$

$$\textcircled{4} \int (\sinh x + \cosh x) dx$$

$$\int \sinh x dx + \int \cosh x dx$$

$$= \cosh x + \sinh x + C$$